Retrieval-Augmented Generation: A Survey

Retrieval-Augmented Generation (RAG) combines information retrieval with text generation. Instead of relying solely on the model's parameters to store knowledge, RAG retrieves relevant documents from an external corpus and conditions generation on those documents. This approach reduces hallucination by grounding answers in real documents and allows the model to access up-to-date information without retraining.

The retrieval step typically uses dense embeddings for semantic search, though sparse methods like BM25 are still valuable for exact keyword matching. A common pattern is hybrid search combining both approaches via Reciprocal Rank Fusion (RRF). The key advantage of RRF is that it works on ranks rather than raw scores, so BM25 scores and cosine similarities do not need to be calibrated to the same numerical scale.

BM25 is a probabilistic relevance model and a cornerstone of modern text retrieval. It improves on TF-IDF through term frequency saturation, controlled by the k1 parameter (typically 1.5), and document length normalization via the b parameter (typically 0.75). The IDF component measures how rare a term is across the collection.

Benchmark Results: Retrieval Strategies

We evaluated four retrieval strategies on the Natural Questions dataset. The table below summarizes the results across three key metrics: Recall@5, MRR, and latency per query. All experiments were conducted on an Apple M2 MacBook Air with 8GB unified memory.

Strategy	Recall@5	MRR	Latency(ms)
BM25 Only	0.72	0.58	12
Vector Only	0.81	0.67	45
Hybrid RRF	0.88	0.74	48
Hybrid+Reranker	0.93	0.82	320

Conclusion

The hybrid approach with reranking achieves the highest accuracy, but at the cost of significantly higher latency. For real-time applications, the base hybrid RRF strategy offers the best trade-off between accuracy and speed. On memory-constrained devices like the M2 MacBook Air, on-demand model loading is essential to stay within the 8GB budget.